

Chrono Delivery

Problem ID : problem-36-22971029
Time limit : 3s

The Chrono Delivery Guild ships parcels between the N space stations of the Outer Reach, which are joined by one-way wormholes, each with a travel time. Some are *temporal* wormholes with negative travel time. The dispatcher Spock must find the minimum total time from station a to station b for each delivery request, using wormholes any number of times. If a route can pass through a *time loop*, a cycle with negative total time, the courier could arrive arbitrarily early, and the request is a paradox.

Input

The first line contains three integers N , M and Q ($1 \leq N \leq 150$, $0 \leq M \leq 10\,000$, $1 \leq Q \leq 100\,000$): the number of stations, the number of wormholes and the number of delivery requests. Stations are numbered 0 to $N - 1$.

Each of the next M lines contains three integers u , v and t ($0 \leq u, v < N$, $-10\,000 \leq t \leq 10\,000$), describing a one-way wormhole from station u to station v with travel time t . There may be several wormholes between the same pair of stations, and a wormhole may lead from a station back to itself.

Each of the next Q lines contains two integers a and b ($0 \leq a, b < N$): a request to travel from station a to station b .

Output

For each request, in the order given, output a single line containing:

- UNREACHABLE if there is no route from a to b ;
- PARADOX if some route from a to b passes through a station that lies on a time loop, so arbitrarily small times are possible;
- otherwise, the minimum total travel time from a to b as an integer (which may be negative).

A courier who does not move takes time 0, so a request with $a = b$ has answer 0 unless it is a paradox.

Note

With up to 100 000 requests, reading input line by line with `input()` may be slow in Python. Consider reading all input at once with `sys.stdin.buffer.read().split()`.

Sample input 1

```
4 4 4
0 1 -2
0 2 1
2 3 -3
3 2 2
0 1
0 3
3 1
1 1
```

Sample output 1

```
-2
PARADOX
UNREACHABLE
0
```

Sample input 2

```
5 5 5
0 1 4
1 2 -2
2 1 1
2 3 3
0 4 7
0 3
0 4
1 1
3 1
0 0
```

Sample output 2

```
PARADOX
7
PARADOX
UNREACHABLE
0
```